

1. Password Reset / Account Lockout

****Symptoms:****

- User cannot log in to workstation, email, or VPN
- Messages such as "Account locked" or "Password expired"
- Repeated failed login attempts shown in security logs

****Root Causes:****

- Multiple failed login attempts (typically 3-5 attempts)
- Password expiry due to policy (90-day cycle)
- Cached or saved credentials on mobile devices/browsers
- Caps Lock or keyboard layout issues
- Account compromise attempts

****Fix Steps:****

1. ****Verify user identity**** according to company policy (employee ID, manager approval, security questions)
2. ****Check account status**** in Active Directory Users and Computers
3. ****Review recent login attempts**** in security logs for suspicious activity
4. ****Unlock account**** if locked, reset password if expired
5. ****Set temporary password**** meeting complexity requirements
6. ****Force password change**** at next login
7. ****Check for service accounts**** using old credentials
8. ****Update mobile devices**** and email clients with new password
9. ****Document incident**** in ticketing system

****Verification:****

- User successfully logs in to workstation
- Email access confirmed
- VPN connection tested

- All mapped drives accessible

****Rollback:****

- Re-lock account if suspicious behavior detected
- Restore previous password if business-critical applications fail
- Document rollback reasons

****Escalation:****

- Repeated lockouts (>3 times/day) → Security team
- Unusual login locations or times → SOC
- Potential brute force indicators → CISO

2. MFA / 2FA Failure

****Symptoms:****

- OTP not received via SMS/email
- Push notification denied or expired
- Authenticator app codes not working
- "Invalid token" error messages

****Root Causes:****

- Time drift on device (>30 seconds difference)
- New or replaced phone without updating MFA
- MFA token desynchronization
- Network connectivity issues
- SIM card problems
- App permissions blocked

****Fix Steps:****

1. ****Verify user identity**** with multi-factor checks
2. ****Check system time**** on user's device (must be within 30 seconds)
3. ****Test network connectivity**** to MFA servers
4. ****Resync MFA token**** in admin console
5. ****Remove and re-enroll**** MFA device if resync fails
6. ****Issue temporary MFA bypass**** only with manager approval
7. ****Check SMS gateway**** status for SMS-based MFA
8. ****Verify phone number**** format in user profile
9. ****Test alternative MFA methods**** if available

****Verification:****

- Successful MFA-authenticated login from user's device
- All MFA methods tested and working
- Backup codes generated if needed

****Rollback:****

- Remove temporary bypass after 24 hours
- Re-enroll previous device if new one fails

****Escalation:****

- Multiple users affected → Identity team
- MFA server issues → Infrastructure team
- Suspected compromise → Security team immediately

3. No Internet Access

****Symptoms:****

- No connectivity indicator (red X or yellow triangle)
- Websites do not load
- "Page cannot be displayed" errors
- Limited connectivity warnings

****Root Causes:****

- DNS or gateway unreachable
- Network adapter driver issues
- DHCP lease expiration
- Proxy configuration problems
- Firewall blocking
- ISP outage

****Fix Steps:****

1. ****Check physical connection**** (cable seated, Wi-Fi switch on)

2. ****Verify Wi-Fi connection**** to correct SSID
3. ****Run network troubleshooter**** (Windows + R, ncpa.cpl)
4. ****Release and renew IP**** (ipconfig /release, ipconfig /renew)
5. ****Flush DNS cache**** (ipconfig /flushdns)
6. ****Test with static DNS**** (8.8.8.8, 1.1.1.1)
7. ****Check proxy settings**** (Internet Options > Connections > LAN)
8. ****Disable/enable network adapter****
9. ****Update network drivers**** from Device Manager
10. ****Check Windows Firewall**** settings

****Verification:****

- Successful browsing to multiple websites
- Ping tests to 8.8.8.8 and internal servers
- All network resources accessible

****Rollback:****

- Restore previous DNS settings if needed
- Re-enable disabled adapters
- Document any configuration changes

****Escalation:****

- Multiple users affected → Network team immediately
- ISP issues → Telecom team
- Hardware failure → Desktop team

4. Slow Computer Performance

****Symptoms:****

- Applications take >30 seconds to open
- System freezes or hangs
- High CPU/memory usage in Task Manager
- Boot time >5 minutes

****Root Causes:****

- Too many startup programs
- Insufficient RAM (usage >80%)
- Hard drive >85% full
- Malware infection
- Outdated drivers
- Windows updates pending

****Fix Steps:****

1. ****Check Task Manager**** for resource hogs (Ctrl+Shift+Esc)

2. ****Review startup programs**** (Task Manager > Startup tab)
3. ****Run Disk Cleanup**** (cleanmgr.exe)
4. ****Check hard drive space**** (must have >15% free)
5. ****Disable unnecessary services**** (msconfig.exe)
6. ****Update Windows**** and all drivers
7. ****Run antivirus scan**** (full system scan)
8. ****Check for Windows updates**** pending restart
9. ****Defragment hard drive**** (if not SSD)
10. ****Check Event Viewer**** for system errors

****Verification:****

- Boot time <2 minutes
- Applications open within 10 seconds
- CPU usage <50% during normal operation
- Memory usage <70%

****Rollback:****

- Re-enable disabled startup programs if needed
- Restore previous power settings
- Document all changes made

****Escalation:****

- Hardware failure suspected → Hardware team
- Malware detected → Security team
- Multiple affected users → Infrastructure

5. Blue Screen of Death (BSOD)

****Symptoms:****

- System crashes with blue screen
- Automatic restart loops
- Specific error codes (0x000000...)
- Memory dump creation

****Root Causes:****

- Driver conflicts or corruption
- Hardware failure (RAM, HDD, motherboard)
- Overheating
- Power supply issues
- Recent Windows updates
- Malware infection

****Fix Steps:****

1. ****Document STOP error code**** and parameters
2. ****Boot into Safe Mode**** (F8 during startup)
3. ****Check Event Viewer**** for critical errors
4. ****Run memory diagnostic**** (mdsched.exe)
5. ****Check disk for errors**** (chkdsk /f /r)
6. ****Update all drivers**** from manufacturer
7. ****Check device manager**** for yellow warnings
8. ****Test with minimal hardware**** (remove peripherals)
9. ****Check system temperatures**** (HWMonitor)
10. ****Run system file checker**** (sfc /scannow)

****Verification:****

- System stable for >4 hours
- No BSOD in Event Viewer
- All hardware recognized in Device Manager

****Rollback:****

- Uninstall problematic Windows updates
- Roll back drivers if needed
- System restore if available

****Escalation:****

- Hardware diagnostics fail → Hardware team
- Persistent BSOD → L3 support
- Multiple systems affected → Infrastructure

6. Printer Not Working

****Symptoms:****

- Print jobs stuck in queue
- "Printer offline" status
- Print quality issues
- Cannot find printer

****Root Causes:****

- Print spooler service stopped
- Driver corruption
- Network connectivity issues
- Paper jams or low supplies
- Wrong default printer
- Permissions issues

****Fix Steps:****

1. ****Check physical printer**** (power, paper, toner)
2. ****Restart print spooler**** (services.msc)
3. ****Clear print queue**** (Control Panel > Devices and Printers)
4. ****Set as default printer**** if needed
5. ****Update/reinstall printer driver****
6. ****Test with different user account****
7. ****Check network connectivity**** (ping printer IP)
8. ****Verify printer permissions**** (Security tab)
9. ****Print test page**** from printer properties
10. ****Check for Windows updates**** affecting printing

****Verification:****

- Test page prints successfully
- Multiple applications can print
- Print quality acceptable

- All users can access printer

****Rollback:****

- Restore previous driver version
- Reconfigure printer settings
- Document any network changes

****Escalation:****

- Network printer issues → Network team
- Server-based printing problems → Server team
- Multiple printers affected → Infrastructure

7. Email Not Sending/Receiving

****Symptoms:****

- Emails stuck in Outbox
- No new emails for extended period
- Send/receive errors
- Attachment upload failures

****Root Causes:****

- Outlook profile corruption
- Exchange server issues
- Mailbox full (>95% capacity)
- Authentication problems
- Add-in conflicts
- Network connectivity

****Fix Steps:****

1. ****Check mailbox size**** (File > Tools > Mailbox Cleanup)
2. ****Test email in Outlook Web App****
3. ****Check send/receive settings**** (Send/Receive groups)
4. ****Repair Outlook profile**** (Control Panel > Mail)
5. ****Disable add-ins**** one by one (File > Options > Add-ins)
6. ****Check cached mode settings****
7. ****Create new Outlook profile**** if repair fails
8. ****Verify Exchange server status**** (ping, telnet)
9. ****Check antivirus**** email scanning settings
10. ****Empty Deleted Items**** and Junk folders

****Verification:****

- Send test email to external address
- Receive test email

- Attachments <25MB work

- Calendar syncing properly

****Rollback:****

- Restore previous Outlook settings

- Re-enable disabled add-ins if needed

- Document profile changes

****Escalation:****

- Exchange server issues → Messaging team

- Multiple users affected → Server team

- Authentication problems → Identity team

8. VPN Connection Issues

****Symptoms:****

- Cannot connect to VPN
- Frequent VPN disconnections
- Slow VPN performance
- Authentication failures

****Root Causes:****

- Incorrect VPN client configuration
- Firewall blocking VPN ports
- Certificate expiration
- Split tunneling issues
- ISP blocking VPN
- Outdated VPN client

****Fix Steps:****

1. ****Verify VPN server address** and credentials**
2. ****Check Windows Firewall** for VPN exceptions**
3. ****Test with Windows built-in VPN** vs client**
4. ****Update VPN client** to latest version**
5. ****Check certificate validity** (if using certs)**
6. ****Disable IPv6** on VPN adapter**
7. ****Try different VPN protocols** (IKEv2, SSTP)**
8. ****Check home router** for VPN passthrough**
9. ****Test from different network** (mobile hotspot)**
10. ****Review VPN logs** for specific errors**

****Verification:****

- Successful VPN connection

- Access to internal resources
- Speed test acceptable (>10 Mbps)
- Connection stable for >30 minutes

****Rollback:****

- Restore previous VPN settings
- Re-enable disabled protocols
- Document configuration changes

****Escalation:****

- VPN server issues → Network team
- Certificate problems → Security team
- Multiple users affected → Infrastructure

9. Software Installation Failures

****Symptoms:****

- "Installation failed" errors
- Insufficient permissions messages
- MSI installer errors
- Application won't launch after install

****Root Causes:****

- Insufficient user permissions
- Corrupted Windows Installer
- Missing dependencies (.NET, Visual C++)
- Antivirus blocking
- Disk space issues

- Previous installation remnants

****Fix Steps:****

1. ****Check user permissions**** (local admin required?)
2. ****Run as administrator**** (right-click installer)
3. ****Check disk space**** (need 2x application size)
4. ****Disable antivirus**** temporarily
5. ****Run Windows Update**** for dependencies
6. ****Install .NET Framework**** if missing
7. ****Clean previous installations**** (Revo Uninstaller)
8. ****Register Windows Installer**** (msiexec /unregister, /regserver)
9. ****Check Event Viewer**** for MSI errors
10. ****Use compatibility mode**** for older software

****Verification:****

- Application launches successfully

- All features functional

- No errors in Event Viewer

- Can uninstall cleanly

****Rollback:****

- Uninstall problematic application

- Restore antivirus settings

- Document any registry changes

****Escalation:****

- Licensing issues → Procurement

- Compatibility problems → Application team

- Multiple failures → Desktop team

10. Windows Update Problems

****Symptoms:****

- Updates fail to install
- System stuck on update screen
- Revert changes on boot
- High disk usage after updates

****Root Causes:****

- Corrupted Windows Update components
- Insufficient disk space
- Conflicting software
- Network interruptions

- System file corruption

****Fix Steps:****

1. ****Check disk space**** (need 10GB+ free)
2. ****Run Windows Update troubleshooter****
3. ****Clear Windows Update cache**** (SoftwareDistribution folder)
4. ****Run SFC scan**** (sfc /scannow)
5. ****Check services**** (Windows Update, BITS running)
6. ****Install updates manually**** from Catalog
7. ****Disable non-Microsoft services**** (clean boot)
8. ****Update device drivers**** first
9. ****Check for pending reboots****
10. ****Reset Windows Update components**** via script

****Verification:****

- All updates install successfully
- System boots normally
- No errors in Windows Update history
- Performance normal

****Rollback:****

- Uninstall problematic updates
- System restore if available
- Document KB numbers removed

****Escalation:****

- Update server issues → Server team
- Multiple systems affected → Infrastructure
- Business-critical application conflicts → Application team

11. Audio/Sound Issues

****Symptoms:****

- No sound from speakers/headphones
- Microphone not working
- Audio crackling or distorted
- Wrong playback device selected

****Root Causes:****

- Wrong audio device set as default
- Driver corruption or outdated
- Hardware muted or disconnected
- Windows audio service stopped

- Application-specific settings

- Hardware failure

****Fix Steps:****

1. ****Check physical connections**** (3.5mm jack, USB)
2. ****Verify volume levels**** (system and application)
3. ****Set correct playback device**** (right-click speaker icon)
4. ****Run Windows audio troubleshooter****
5. ****Update audio drivers**** (Device Manager)
6. ****Check default format**** (24-bit, 48000 Hz)
7. ****Test with different applications****
8. ****Disable audio enhancements****
9. ****Restart Windows Audio service****
10. ****Check BIOS**** for onboard audio settings

****Verification:****

- Audio plays from multiple sources
- Microphone records clearly
- No distortion at normal volume
- Headset/speaker switching works

****Rollback:****

- Restore previous audio settings
- Re-enable disabled enhancements
- Document driver versions

****Escalation:****

- Hardware failure → Hardware team
- Driver conflicts → Desktop team
- Multiple users → Infrastructure

12. USB Device Not Recognized

****Symptoms:****

- "USB device not recognized" popup
- Device shows in Device Manager with error
- Device works intermittently
- No power to USB device

****Root Causes:****

- Insufficient power (USB hub overloaded)
- Driver issues or corruption
- USB port damage

- Windows power management

- Device compatibility

- USB selective suspend

****Fix Steps:****

1. ****Try different USB ports**** (USB 2.0 vs 3.0)
2. ****Check Device Manager**** for error codes
3. ****Update USB drivers**** (chipset drivers)
4. ****Disable USB selective suspend**** (Power Options)
5. ****Uninstall device**** from Device Manager, restart
6. ****Check power requirements**** (avoid unpowered hubs)
7. ****Update BIOS**** and chipset drivers
8. ****Test on different computer****
9. ****Check USB cable**** (try different cable)
10. ****Run Hardware troubleshooter****

****Verification:****

- Device recognized consistently
- Data transfer works
- No errors in Device Manager
- Device survives reboot

****Rollback:****

- Re-enable power management
- Restore previous drivers
- Document BIOS version changes

****Escalation:****

- Hardware damage → Hardware team
- Driver conflicts → Desktop team

- Multiple devices → Infrastructure

13. Screen Display Issues

****Symptoms:****

- Wrong screen resolution
- Multiple monitors not detected
- Screen flickering
- Black screen after login

****Root Causes:****

- Graphics driver issues
- Wrong display settings

- Cable connection problems
- Graphics card failure
- Windows display scaling
- Monitor compatibility

****Fix Steps:****

1. ****Check cable connections**** (HDMI, DisplayPort, VGA)
2. ****Detect displays**** in Windows settings
3. ****Update graphics drivers**** (Intel, NVIDIA, AMD)
4. ****Set recommended resolution**** for each monitor
5. ****Check refresh rate**** (60Hz minimum)
6. ****Test with different cable types****
7. ****Run Windows Display troubleshooter****
8. ****Check graphics card seating**** (desktop)
9. ****Adjust display scaling**** (100%, 125%, 150%)

10. **Test with onboard graphics if available**

****Verification:****

- All monitors detected at correct resolution
- No flickering or artifacts
- Extended desktop works properly
- Survives reboot and sleep

****Rollback:****

- Restore previous resolution settings
- Roll back graphics drivers
- Document scaling changes

****Escalation:****

- Hardware failure → Hardware team

- Multiple monitors → Desktop team
- Graphics card issues → Hardware team

14. Battery/Power Issues (Laptops)

****Symptoms:****

- Battery not charging
- Laptop shuts down unexpectedly
- Battery drains quickly
- No power when plugged in

****Root Causes:****

- Battery wear/degradation

- Power adapter failure
- Charging port damage
- Power management settings
- BIOS issues
- Thermal throttling

****Fix Steps:****

1. ****Check power adapter**** (LED indicator, different outlet)
2. ****Inspect charging port**** for damage/debris
3. ****Remove battery**** (if removable), clean contacts
4. ****Update BIOS**** and power management drivers
5. ****Calibrate battery**** (full discharge/recharge)
6. ****Check power plan settings**** (balanced vs high performance)
7. ****Run battery report**** (powercfg /batteryreport)
8. ****Test with different charger**** (same voltage/watts)

9. ****Check for overheating**** (clean vents)

10. ****Disable fast startup**** temporarily

****Verification:****

- Battery charges to 100%
- Holds charge for expected time
- No unexpected shutdowns
- Power adapter recognized

****Rollback:****

- Restore previous power plans
- Re-enable fast startup
- Document BIOS changes

****Escalation:****

- Hardware failure → Hardware team

- Multiple laptops → Procurement

- Safety concerns → Facilities

15. Touchpad/Keyboard Issues

****Symptoms:****

- Touchpad not responding

- Keyboard typing wrong characters

- Some keys not working

- Touchpad too sensitive

****Root Causes:****

- Driver corruption or outdated
- Hardware disable key pressed
- Liquid damage
- Settings misconfiguration
- Windows updates affecting drivers

****Fix Steps:****

1. ****Check function key**** for touchpad toggle (Fn+F5/F7)
2. ****Check Device Manager**** for driver issues
3. ****Update touchpad/keyboard drivers****
4. ****Check touchpad settings**** (sensitivity, palm rejection)
5. ****Test in BIOS**** (rules out Windows issues)
6. ****Run Hardware troubleshooter****
7. ****Check for stuck keys**** (keyboard test)
8. ****Update BIOS**** and chipset drivers

9. ****Test with external mouse/keyboard****

10. ****Check language settings** (wrong keyboard layout)**

****Verification:****

- All keys respond correctly
- Touchpad gestures work
- Sensitivity appropriate
- Survives reboot

****Rollback:****

- Restore previous touchpad settings
- Revert keyboard layout changes
- Document driver versions

****Escalation:****

- Hardware damage → Hardware team

- Multiple systems → Desktop team

- Liquid damage → Facilities

16. Windows Hello/Fingerprint Issues

****Symptoms:****

- Fingerprint not recognized

- Windows Hello PIN not working

- Facial recognition fails

- "This option is currently unavailable"

****Root Causes:****

- Biometric driver issues
- Windows Hello service problems
- Hardware not compatible
- Group Policy restrictions
- User profile corruption

****Fix Steps:****

1. ****Check biometric devices**** in Device Manager
2. ****Update biometric drivers**** (fingerprint, camera)
3. ****Run Windows Hello troubleshooter****
4. ****Check Group Policy**** (gpedit.msc > Biometrics)
5. ****Remove and re-enroll**** biometric data
6. ****Check Windows Biometric Service**** (running)
7. ****Clear TPM**** (if PIN issues)
8. ****Create new Windows Hello PIN****

9. ****Check for Windows updates****

10. ****Test with different user account****

****Verification:****

- Fingerprint enrolls and recognizes
- PIN works consistently
- Facial recognition works in good lighting
- All Hello methods functional

****Rollback:****

- Restore previous biometric settings
- Re-enable disabled services
- Document TPM changes

****Escalation:****

- Hardware failure → Hardware team
- Group Policy issues → Security team
- Multiple users → Identity team

17. OneDrive Sync Issues

****Symptoms:****

- Files not syncing
- OneDrive sync errors
- "Processing changes" stuck
- Files show sync pending icon

****Root Causes:****

- OneDrive service issues
- File path too long (>260 characters)
- Special characters in filenames
- Insufficient storage space
- Network connectivity
- File locked by application

****Fix Steps:****

1. ****Check OneDrive service status**** online
2. ****Check storage quota**** (need 10% free)
3. ****Pause and resume sync****
4. ****Reset OneDrive**** (OneDrive.exe /reset)
5. ****Check for long file paths**** (>260 chars)
6. ****Remove special characters**** from filenames
7. ****Check file permissions**** (must have modify)

8. ****Clear OneDrive cache**** (AppData\Local\Microsoft\OneDrive)

9. ****Check for locked files**** (close Office apps)

10. ****Update OneDrive**** to latest version

****Verification:****

- Files sync within 5 minutes
- No sync errors in activity center
- All folders up to date
- SharePoint sites sync properly

****Rollback:****

- Restore previous sync settings
- Reconfigure selective sync
- Document folder changes

****Escalation:****

- OneDrive service issues → Microsoft support
- SharePoint sync issues → SharePoint team
- Multiple users → Infrastructure

18. Teams/Zoom Audio Video Issues

****Symptoms:****

- Microphone not working in Teams/Zoom
- Camera not detected
- Poor call quality
- Echo or feedback

****Root Causes:****

- App permissions blocked
- Wrong device selected in app
- Driver issues
- Network bandwidth
- Audio enhancements
- Conflicting applications

****Fix Steps:****

1. ****Check app permissions**** (Privacy > Microphone/Camera)
2. ****Select correct devices**** in Teams/Zoom settings
3. ****Test audio/video**** in app settings
4. ****Update audio/video drivers****
5. ****Check Windows privacy settings****
6. ****Close other apps**** using camera/mic

7. ****Disable audio enhancements****

8. ****Check network bandwidth**** (need 1.5Mbps up/down)

9. ****Try web version**** vs desktop app

10. ****Clear app cache**** (%appdata%\Microsoft\Teams)

****Verification:****

- Audio clear without echo
- Video shows properly
- No network quality warnings
- Multiple participants can hear/see

****Rollback:****

- Restore previous app settings
- Re-enable audio enhancements
- Document permission changes

****Escalation:****

- Network issues → Network team
- Multiple users → Infrastructure
- Licensing issues → Procurement

19. Certificate Errors in Browser

****Symptoms:****

- "Your connection is not private"
- Certificate warnings on websites
- "NET::ERR_CERT_DATE_INVALID"
- Company websites show as insecure

****Root Causes:****

- System date/time incorrect
- Missing root certificates
- Antivirus SSL scanning
- Proxy configuration
- Expired certificates
- Browser cache issues

****Fix Steps:****

1. ****Check system date/time**** (must be accurate)
2. ****Clear browser cache**** and cookies
3. ****Update root certificates**** (Windows Update)
4. ****Check antivirus**** SSL scanning settings
5. ****Import company certificates**** if needed

6. ****Check proxy settings**** (automatic vs manual)

7. ****Try different browser****

8. ****Check certificate store**** (certmgr.msc)

9. ****Disable browser extensions**** temporarily

10. ****Run Windows Update**** for root certificates

****Verification:****

- No certificate warnings on sites

- Company sites show as secure

- All browsers work properly

- Internal sites accessible

****Rollback:****

- Restore previous browser settings

- Re-enable SSL scanning if needed

- Document certificate imports

****Escalation:****

- Company certificate issues → Security team
- Multiple users → Infrastructure
- Proxy issues → Network team

20. Remote Desktop Connection Issues

****Symptoms:****

- Cannot connect to remote computer
- "Remote Desktop can't find the computer"
- Authentication failures

- **Black screen after connection**

****Root Causes:****

- **Remote Desktop not enabled**
- **Firewall blocking RDP**
- **Network connectivity issues**
- **Wrong credentials**
- **RDP service issues**
- **Group Policy restrictions**

****Fix Steps:****

1. ****Check if RDP is enabled** (System > Remote Settings)**
2. ****Verify computer name/IP** is correct**
3. ****Check Windows Firewall** for RDP exceptions**
4. ****Test with IP address** vs computer name**

5. ****Check if user is in Remote Desktop Users**** group
6. ****Verify network connectivity**** (ping, port 3389)
7. ****Check RDP service**** (TermService running)
8. ****Try different RDP client****
9. ****Check Group Policy**** (gpresult /r)
10. ****Review RDP logs**** (Event Viewer)

****Verification:****

- Successful connection established
- Desktop appears properly
- Can transfer files if enabled
- Connection stable

****Rollback:****

- Restore previous RDP settings

- Re-enable firewall if needed
- Document user group changes

****Escalation:****

- Network connectivity → Network team
- Group Policy issues → Security team
- Server-side problems → Server team

21. Windows Search Not Working

****Symptoms:****

- Start menu search shows no results
- File Explorer search fails

- Search is very slow
- Cortana not responding

****Root Causes:****

- Windows Search service stopped
- Index corruption
- Indexing not complete
- File location not indexed
- Windows updates issues

****Fix Steps:****

1. ****Restart Windows Search service**** (services.msc)
2. ****Rebuild search index**** (Control Panel > Indexing Options)
3. ****Check indexed locations**** (add needed folders)
4. ****Run Search troubleshooter****

5. ****Check disk space**** (need 15% free for indexing)

6. ****Restart Cortana**** (Task Manager)

7. ****Clear search history**** (Settings > Search)

8. ****Check file types**** in indexing options

9. ****Disable antivirus**** temporarily for indexing

10. ****Run Windows Update**** for search fixes

****Verification:****

- Start menu finds applications
- File search works in Explorer
- Search completes within 10 seconds
- Cortana responds to queries

****Rollback:****

- Restore previous indexing locations

- Re-enable disabled services

- Document index rebuild

****Escalation:****

- Multiple users → Infrastructure

- Enterprise search issues → SharePoint team

- Performance problems → Desktop team

22. Windows Store/App Issues

****Symptoms:****

- Windows Store won't open

- Apps won't download/update

- "Something happened" errors

- Apps crash on startup

****Root Causes:****

- Windows Store cache corruption

- Date/time incorrect

- Microsoft account issues

- Proxy/firewall blocking

- App permissions

- Windows Update missing

****Fix Steps:****

1. ****Clear Store cache**** (wsreset.exe)

2. ****Check date/time settings****

3. ****Sign out/in Microsoft account****

4. ****Check proxy/firewall** for Store URLs**
5. ****Run Windows Store troubleshooter****
6. ****Check app permissions** (Privacy settings)**
7. ****Re-register Store** (PowerShell command)**
8. ****Update Windows** fully**
9. ****Create new user profile** to test**
10. ****Check Storage settings** (apps need space)**

****Verification:****

- Store opens and loads
- Apps download/install
- Updates work properly
- No crashes in apps

****Rollback:****

- Restore previous Store settings

- Re-register apps if needed

- Document account changes

****Escalation:****

- Licensing issues → Procurement

- Multiple users → Infrastructure

- Microsoft account → Identity team

23. Windows Defender/Antivirus Issues

****Symptoms:****

- Real-time protection won't turn on

- Scanning fails
- Update errors
- False positive detections

****Root Causes:****

- Third-party antivirus conflicts
- Definition updates corrupted
- System file corruption
- Group Policy restrictions
- Malware blocking antivirus

****Fix Steps:****

1. ****Check for conflicting antivirus**** (McAfee, Norton)
2. ****Update virus definitions**** manually
3. ****Run Windows Update**** for Defender updates

4. ****Check Group Policy**** (gpedit.msc)
5. ****Run offline scan**** (Windows Defender Offline)
6. ****Check services**** (Windows Defender services running)
7. ****Run SFC scan**** (sfc /scannow)
8. ****Check Event Viewer**** for errors
9. ****Add exclusions**** if false positives
10. ****Check Windows Security**** settings

****Verification:****

- Real-time protection active
- Full scan completes
- Definitions up to date
- No false positives

****Rollback:****

- Remove added exclusions
- Restore previous settings
- Document policy changes

****Escalation:****

- Malware detected → Security team
- Multiple systems → Infrastructure
- Policy issues → Security team

24. Time/Date Sync Issues

****Symptoms:****

- System time wrong

- "Your clock is behind/ahead" errors

- Time zone wrong

- Time changes after reboot

****Root Causes:****

- CMOS battery failing

- Wrong time zone

- Time sync service issues

- Domain controller problems

- Manual time settings

****Fix Steps:****

1. ****Set correct time zone****

2. ****Sync with internet time**** (time.windows.com)

3. ****Check Windows Time service**** (running)

4. ****Check CMOS battery**** (if time resets)
5. ****Sync with domain controller**** (domain joined)
6. ****Check registry**** for time sync settings
7. ****Run W32tm /resync**** command
8. ****Check Group Policy**** for time settings
9. ****Update BIOS**** if battery issues
10. ****Configure NTP server**** if needed

****Verification:****

- Time accurate to within 30 seconds
- Survives reboot
- Correct time zone
- Syncs automatically

****Rollback:****

- Restore previous time settings
- Reconfigure NTP if changed
- Document battery replacement

****Escalation:****

- Domain time issues → Server team
- Multiple systems → Infrastructure
- CMOS battery → Hardware team

25. Network Share Access Issues

****Symptoms:****

- "Access denied" to network drives

- Cannot connect to shared folders

- "Network path not found"

- Credentials not accepted

****Root Causes:****

- Wrong credentials cached

- Permissions changed

- Network connectivity

- DNS resolution issues

- Workstation trust issues

****Fix Steps:****

1. ****Check network connectivity**** (ping server)

2. ****Clear cached credentials**** (Credential Manager)

3. ****Check share permissions**** (effective access)

4. ****Test with IP address** vs server name**
5. ****Check DNS resolution** (nslookup)**
6. ****Rejoin domain** if trust issues**
7. ****Map drive with different credentials****
8. ****Check time sync** (Kerberos needs time sync)**
9. ****Test from different computer****
10. ****Check Event Viewer** for logon errors**

****Verification:****

- Can access all permitted shares
- Credentials work consistently
- No access denied errors
- Speed acceptable

****Rollback:****

- Remap previous drives
- Restore cached credentials
- Document permission changes

****Escalation:****

- Server issues → Server team
- Permissions problems → Security team
- Multiple users → Infrastructure

26. Outlook Performance Issues

****Symptoms:****

- Outlook very slow to respond

- Search takes forever
- Freezes when composing
- High CPU usage

****Root Causes:****

- Large mailbox size
- Too many add-ins
- Cached mode issues
- Corrupted OST file
- Antivirus scanning
- Network latency

****Fix Steps:****

1. ****Check mailbox size**** (File > Tools > Mailbox Cleanup)
2. ****Disable unnecessary add-ins****

3. ****Compact OST file**** (File > Account Settings)
4. ****Rebuild search index**** for Outlook
5. ****Check cached mode settings**** (disable temporarily)
6. ****Run ScanPST**** on PST files if used
7. ****Archive old emails**** to reduce size
8. ****Check antivirus**** Outlook integration
9. ****Create new Outlook profile****
10. ****Check network performance**** to Exchange

****Verification:****

- Outlook opens within 10 seconds
- Search completes quickly
- No freezing during use
- CPU usage normal

****Rollback:****

- Re-enable disabled add-ins
- Restore cached mode
- Document profile changes

****Escalation:****

- Exchange issues → Messaging team
- Multiple users → Infrastructure
- Add-in problems → Application team

27. Wi-Fi Connection Problems

****Symptoms:****

- Cannot see Wi-Fi networks

- "Unable to connect" errors

- Limited connectivity

- Keeps disconnecting

****Root Causes:****

- Wrong Wi-Fi drivers

- Network profile corrupted

- Security type mismatch

- Power saving settings

- Router issues

- Signal interference

****Fix Steps:****

1. ****Update Wi-Fi drivers**** (manufacturer website)

2. ****Forget and re-add** network profile**
3. ****Check security type** (WPA2 vs WPA3)**
4. ****Disable power saving** for Wi-Fi adapter**
5. ****Run Network troubleshooter****
6. ****Check router firmware** (contact network team)**
7. ****Change Wi-Fi channel** (if admin)**
8. ****Disable IPv6** temporarily**
9. ****Check for Windows updates****
10. ****Test with mobile hotspot****

****Verification:****

- Connects automatically
- Stable connection >30 minutes
- Good signal strength
- Expected speeds

****Rollback:****

- Restore previous drivers
- Re-enable power management
- Document security changes

****Escalation:****

- Router/access point → Network team
- Multiple users → Infrastructure
- Driver issues → Desktop team

28. Windows Activation Issues

****Symptoms:****

- "Activate Windows" watermark
- "Windows is not genuine"
- Activation fails with error
- Product key not accepted

****Root Causes:****

- Hardware changes (motherboard)
- Invalid product key
- KMS server issues
- Blocked volume license key
- System file corruption

****Fix Steps:****

1. ****Check current activation status**** (slmgr /dli)

2. ****Try automatic activation**** online
3. ****Check product key**** (if OEM, should auto-activate)
4. ****Run activation troubleshooter****
5. ****Check KMS server**** (domain environment)
6. ****Re-enter product key**** (slmgr /ipk)
7. ****Check system files**** (sfc /scannow)
8. ****Verify hardware**** (major changes need reactivation)
9. ****Contact Microsoft support**** if OEM issues
10. ****Check Event Viewer**** for activation errors

****Verification:****

- Activation successful
- No watermark
- All features work
- Survives reboot

****Rollback:****

- Document previous key if changed
- Restore KMS settings
- Note hardware changes

****Escalation:****

- Volume licensing → Procurement
- KMS server issues → Server team
- Multiple systems → Infrastructure

29. File Recovery/Deleted Files

****Symptoms:****

- Accidentally deleted important files
- Files missing after cleanup
- Emails deleted permanently
- Need previous version of file

****Root Causes:****

- User error
- Cleanup utilities
- Storage space issues
- Sync errors
- Malicious deletion

****Fix Steps:****

1. ****Check Recycle Bin**** immediately

2. ****Check OneDrive recycle bin**** (30 days)
3. ****Check email deleted items**** (and recoverable items)
4. ****Check Previous Versions**** (right-click > Properties)
5. ****Check backup system**** (Veeam, etc.)
6. ****Stop using drive**** to prevent overwriting
7. ****Check shadow copies**** (vssadmin list shadows)
8. ****Contact IT immediately**** for critical files
9. ****Document what/when deleted****
10. ****Check sync conflicts**** (OneDrive)

****Verification:****

- Files recovered intact
- Open properly in application
- No corruption
- All data present

****Rollback:****

- Document recovery method
- Note any data loss
- Update backup procedures

****Escalation:****

- Critical business data → Manager
- Legal/regulatory → Compliance
- No backups available → Infrastructure

30. System Restore/Recovery

****Symptoms:****

- Need to undo system changes
- System unstable after updates
- Want to go back to previous state
- Recovery partition not working

****Root Causes:****

- Bad Windows updates
- Driver issues
- Software conflicts
- Registry corruption
- System changes

****Fix Steps:****

1. ****Check restore points**** available (rstrui.exe)

2. ****Create new restore point**** before changes
3. ****Boot to Safe Mode**** if needed
4. ****Run System Restore**** from recovery
5. ****Check recovery partition**** (manufacturer)
6. ****Use installation media**** if recovery fails
7. ****Check disk space**** for restore points
8. ****Configure restore point**** settings
9. ****Test recovery options**** before needed
10. ****Document restore point**** creation schedule

****Verification:****

- System boots normally
- Previous functionality restored
- No data loss
- Applications work

****Rollback:****

- Revert restore if issues
- Document what was undone
- Note lessons learned

****Escalation:****

- System won't boot → Hardware team
- Data recovery needed → Specialist
- Multiple systems → Infrastructure

31. Performance Monitoring/Troubleshooting

****Symptoms:****

- System running slow
- Need to identify bottlenecks
- High resource usage
- Performance degradation over time

****Root Causes:****

- Memory leaks
- Disk bottlenecks
- CPU intensive processes
- Network latency
- Background tasks

****Fix Steps:****

1. ****Open Task Manager**** (Ctrl+Shift+Esc)

2. ****Check Performance tab**** for resource usage
3. ****Identify high-usage processes****
4. ****Check Startup impact**** ratings
5. ****Use Resource Monitor**** for detailed analysis
6. ****Check Event Viewer**** for errors
7. ****Run Performance troubleshooter****
8. ****Check for Windows updates****
9. ****Analyze boot performance**** (Event Viewer)
10. ****Create performance baseline****

****Verification:****

- CPU usage <80% normally
- Memory usage <80%
- Disk usage <100% sustained
- Boot time <2 minutes

****Rollback:****

- Document changes made
- Restore disabled items if needed
- Note baseline metrics

****Escalation:****

- Hardware bottlenecks → Procurement
- Application issues → Application team
- Multiple systems → Infrastructure

32. Windows Security/Permission Issues

****Symptoms:****

- "Access denied" errors
- Cannot modify files
- UAC prompts for everything
- Cannot install software

****Root Causes:****

- Wrong user permissions
- File ownership issues
- UAC settings too high
- Group Policy restrictions
- Corrupted user profile

****Fix Steps:****

1. ****Check user account type**** (admin vs standard)

2. ****Take ownership**** of files if needed
3. ****Check NTFS permissions**** (effective access)
4. ****Adjust UAC settings**** if appropriate
5. ****Run as administrator**** when needed
6. ****Check Group Policy**** restrictions
7. ****Verify domain membership**** if applicable
8. ****Check credential manager**** for stored creds
9. ****Test with different user account****
10. ****Check file attributes**** (read-only, system)

****Verification:****

- Can access needed resources
- UAC prompts appropriate level
- Permissions work as expected
- No excessive restrictions

****Rollback:****

- Restore previous permissions
- Document ownership changes
- Note policy modifications

****Escalation:****

- Domain issues → Server team
- Policy problems → Security team
- File server → Server team

33. Backup/Recovery Issues

****Symptoms:****

- Backup failing
- Restore not working
- Backup files corrupted
- Scheduled backups not running

****Root Causes:****

- Insufficient storage space
- Network connectivity
- Permission issues
- Backup service problems
- File locks

****Fix Steps:****

1. ****Check backup storage**** space

2. ****Verify backup service** is running**
3. ****Check backup logs** for errors**
4. ****Test manual backup** to isolate**
5. ****Check file permissions** on source/destination**
6. ****Verify network connectivity** to backup location**
7. ****Check for open files** (backup might skip)**
8. ****Update backup software****
9. ****Test restore process** regularly**
10. ****Document backup procedures****

****Verification:****

- Backups complete successfully
- Restore works properly
- No errors in logs
- Data integrity verified

****Rollback:****

- Restore previous backup config
- Document any changes
- Test rollback procedures

****Escalation:****

- Enterprise backup → Backup team
- Multiple failures → Infrastructure
- Data recovery → Specialist

34. Windows Hello Business/PIN Issues

****Symptoms:****

- PIN not working
- Cannot setup Windows Hello
- "This PIN isn't working" error
- Fingerprint enrollment fails

****Root Causes:****

- TPM issues
- PIN complexity requirements
- Biometric hardware problems
- Group Policy settings
- User profile corruption

****Fix Steps:****

1. ****Check TPM status**** (tpm.msc)

2. ****Clear TPM** if issues (carefully)**
3. ****Check PIN requirements** (complexity)**
4. ****Update biometric drivers****
5. ****Check Group Policy** for Hello settings**
6. ****Create new PIN** meeting requirements**
7. ****Check Windows Hello service****
8. ****Update Windows** fully**
9. ****Check Event Viewer** for Hello errors**
10. ****Test with local vs domain account****

****Verification:****

- PIN works consistently
- Biometric enrollment successful
- Hello options available
- No errors reported

****Rollback:****

- Restore previous PIN if possible
- Document TPM changes
- Note policy modifications

****Escalation:****

- TPM hardware issues → Hardware team
- Domain issues → Server team
- Multiple users → Identity team

35. Microsoft Office Issues

****Symptoms:****

- Office apps won't start
- Crashes during use
- Features not working
- Activation problems

****Root Causes:****

- Add-in conflicts
- Corrupted installation
- Licensing issues
- Updates needed
- Profile corruption

****Fix Steps:****

1. ****Run Office repairs**** (Quick then Online)

2. ****Disable add-ins**** one by one
3. ****Update Office**** fully
4. ****Check activation status**** (File > Account)
5. ****Create new Outlook profile**** if email issues
6. ****Run SaRA tool**** (Microsoft Support and Recovery)
7. ****Check for Windows updates****
8. ****Repair corrupted files**** (Office Upload Center)
9. ****Clear Office cache**** (AppData\Local\Microsoft)
10. ****Reinstall Office**** if repairs fail

****Verification:****

- All Office apps start
- No crashes during use
- Features work properly
- Activation successful

****Rollback:****

- Re-enable add-ins
- Restore previous settings
- Document reinstall

****Escalation:****

- Licensing issues → Procurement
- Multiple users → Infrastructure
- SharePoint issues → SharePoint team

36. SharePoint/OneDrive Sync Problems

****Symptoms:****

- Files not syncing
- Sync errors
- Conflicts reported
- SharePoint sites not accessible

****Root Causes:****

- Authentication issues
- File path too long
- Special characters
- Sync conflicts
- Permissions problems

****Fix Steps:****

1. ****Check SharePoint permissions****

2. ****Clear OneDrive cache****
3. ****Check file paths** (limit 260 chars)**
4. ****Remove special characters** from names**
5. ****Resolve sync conflicts** manually**
6. ****Re-authenticate** OneDrive**
7. ****Check storage quotas****
8. ****Update OneDrive** client**
9. ****Check for locked files****
10. ****Reset OneDrive sync****

****Verification:****

- Files sync within 5 minutes
- No sync errors
- All sites accessible
- No conflicts

****Rollback:****

- Restore previous sync settings
- Reconfigure selective sync
- Document changes

****Escalation:****

- SharePoint server → SharePoint team
- Multiple users → Infrastructure
- Permissions → Security team

37. Teams Meeting/Camera Issues

****Symptoms:****

- Camera not working in Teams
- Microphone issues in meetings
- Screen sharing fails
- Meeting join problems

****Root Causes:****

- Device permissions
- Wrong device selected
- Driver issues
- Network bandwidth
- Teams cache corruption

****Fix Steps:****

1. ****Check device permissions**** in Windows

2. ****Select correct devices**** in Teams settings
3. ****Update camera/mic drivers****
4. ****Clear Teams cache**** (%appdata%\Microsoft\Teams)
5. ****Check network bandwidth****
6. ****Test devices**** in Teams settings
7. ****Run Windows camera troubleshooter****
8. ****Check for Windows updates****
9. ****Try web version**** of Teams
10. ****Update Teams**** application

****Verification:****

- Camera works in Teams
- Audio clear
- Screen sharing works
- Can join meetings

****Rollback:****

- Restore previous Teams settings
- Re-enable disabled devices
- Document changes

****Escalation:****

- Network issues → Network team
- Multiple users → Infrastructure
- Licensing → Procurement

38. Browser Issues (Chrome/Edge/IE)

****Symptoms:****

- Browser won't start
- Pages won't load
- Crashes frequently
- Settings won't save

****Root Causes:****

- Corrupted cache/cookies
- Malicious extensions
- Outdated browser
- Proxy settings
- Security software conflicts

****Fix Steps:****

1. ****Clear cache and cookies****

2. ****Disable extensions** one by one**
3. ****Update browser** to latest**
4. ****Check proxy settings****
5. ****Reset browser settings****
6. ****Try InPrivate/Incognito mode****
7. ****Check for malware****
8. ****Disable hardware acceleration****
9. ****Create new browser profile****
10. ****Reinstall browser** if needed**

****Verification:****

- Browser starts normally
- Pages load quickly
- No crashes
- Settings persist

****Rollback:****

- Restore bookmarks if needed
- Re-enable safe extensions
- Document changes

****Escalation:****

- Enterprise browser → Security team
- Multiple users → Infrastructure
- Proxy issues → Network team

39. Windows Update Download/Install Stuck

****Symptoms:****

- Updates download forever
- Installation stuck at X%
- "Preparing to install" loops
- Revert changes on boot

****Root Causes:****

- Corrupted update cache
- Insufficient disk space
- Conflicting software
- Network interruptions
- System file corruption

****Fix Steps:****

1. ****Check disk space**** (need 10GB+)

2. ****Clear update cache**** (SoftwareDistribution folder)
3. ****Run Windows Update troubleshooter****
4. ****Check for pending reboots****
5. ****Disable non-Microsoft services**** (clean boot)
6. ****Run SFC scan**** (sfc /scannow)
7. ****Check Windows Update service****
8. ****Install updates manually**** from Catalog
9. ****Check Event Viewer**** for errors
10. ****Reset Windows Update components****

****Verification:****

- Updates install successfully
- System boots normally
- No errors in history
- All updates applied

****Rollback:****

- Uninstall problematic updates
- System restore if needed
- Document KB numbers

****Escalation:****

- WSUS issues → Server team
- Multiple systems → Infrastructure
- Business app conflicts → Application team

40. High CPU/Memory Usage by System Processes

****Symptoms:****

- System using 50%+ CPU constantly
- Memory usage very high
- System process high usage
- Computer very slow

****Root Causes:****

- Windows Update scanning
- Superfetch/Sysmain service
- Driver memory leaks
- Hardware interrupts
- System file corruption
- Background tasks

****Fix Steps:****

1. ****Check Task Manager**** for details
2. ****Identify specific process**** causing usage
3. ****Check Windows Update**** status
4. ****Disable Superfetch/Sysmain**** temporarily
5. ****Update all drivers****
6. ****Check for hardware issues****
7. ****Run memory diagnostic****
8. ****Check Event Viewer**** for errors
9. ****Disable Windows Search**** temporarily
10. ****Check for malware**** (rootkits)

****Verification:****

- CPU usage <20% idle
- Memory usage normal
- System responsive

- No excessive usage

****Rollback:****

- Re-enable disabled services
- Restore previous settings
- Document changes

****Escalation:****

- Hardware issues → Hardware team
- Driver problems → Desktop team
- Multiple systems → Infrastructure

41. Windows Defender Firewall Issues

****Symptoms:****

- Can't turn on firewall
- Apps blocked incorrectly
- Network discovery issues
- Can't ping computer

****Root Causes:****

- Third-party firewall conflicts
- Group Policy restrictions
- Service not running
- Corrupted firewall rules
- Network profile issues

****Fix Steps:****

1. ****Check Windows Defender Firewall service****
2. ****Check Group Policy** settings**
3. ****Restore default firewall settings****
4. ****Check network profile** (private vs public)**
5. ****Allow apps** through firewall**
6. ****Check for conflicts** (McAfee, Norton)**
7. ****Reset firewall rules** (netsh advfirewall reset)**
8. ****Check Event Viewer** for blocks**
9. ****Create specific allow rules****
10. ****Check advanced settings****

****Verification:****

- Firewall starts properly
- Apps work as expected
- Network discovery works

- Logging shows activity

****Rollback:****

- Restore previous rules

- Document changes made

- Export/import settings

****Escalation:****

- Policy issues → Security team
- Multiple systems → Infrastructure
- Enterprise firewall → Network team

42. BitLocker Encryption Issues

****Symptoms:****

- BitLocker won't turn on
- Recovery key not working
- Encryption paused
- Performance issues with BitLocker

****Root Causes:****

- No TPM or TPM disabled
- Group Policy restrictions
- Disk configuration issues
- Hardware changes
- Recovery key lost

****Fix Steps:****

1. ****Check TPM status** and version**
2. ****Enable TPM** in BIOS if needed**
3. ****Check Group Policy** for BitLocker settings**
4. ****Prepare drive** for BitLocker (MBR/GPT)**
5. ****Save recovery key** to AD or file**
6. ****Check disk for errors** (chkdsk)**

7. ****Update TPM drivers****

8. ****Check BitLocker service****

9. ****Encrypt without TPM** if allowed**

10. ****Document recovery key** securely**

****Verification:****

- Encryption completes
- Recovery key works
- Performance acceptable
- Key backed up

****Rollback:****

- Decrypt if necessary
- Document key locations
- Note policy changes

****Escalation:****

- TPM hardware issues → Hardware team
- Policy questions → Security team
- Recovery key issues → Security team

43. Windows Hello Face/Fingerprint Setup Issues

****Symptoms:****

- Face setup fails
- Fingerprint not enrolling
- "Couldn't recognize you" errors
- Windows Hello options grayed out

****Root Causes:****

- Hardware not compatible
- Drivers missing
- Privacy settings blocking
- Poor lighting conditions
- Biometric database corrupted

****Fix Steps:****

1. ****Check hardware compatibility**** (Hello compatible)
2. ****Update biometric drivers****
3. ****Check privacy settings**** (allow apps)
4. ****Improve lighting**** for face setup
5. ****Clean fingerprint reader****
6. ****Check Windows Hello service****

7. ****Clear biometric data**** and retry
8. ****Check Group Policy**** if domain joined
9. ****Run Windows Hello troubleshooter****
10. ****Test with different user****

****Verification:****

- Face enrolls successfully
- Fingerprint works consistently
- Recognition fast and accurate
- Multiple angles work

****Rollback:****

- Remove biometric data
- Use PIN instead
- Document changes

****Escalation:****

- Hardware compatibility → Hardware team
- Driver issues → Desktop team
- Policy restrictions → Security team

44. Windows Snap/Virtual Desktop Issues

****Symptoms:****

- Snap Assist not working
- Virtual desktops won't create
- Keyboard shortcuts fail
- Aero Peek not working

****Root Causes:****

- **Performance options disabled**
- **Graphics drivers**
- **Registry settings**
- **Group Policy restrictions**
- **System performance**

****Fix Steps:****

1. ****Check Performance Options** (Visual Effects)**
2. ****Update graphics drivers****
3. ****Enable Snap Assist** in Settings**
4. ****Check keyboard shortcuts** (Windows+Tab)**
5. ****Check Group Policy** if applicable**
6. ****Run Performance troubleshooter****

7. ****Check registry**** for Aero settings

8. ****Adjust for best appearance****

9. ****Check display scaling****

10. ****Restart Explorer process****

****Verification:****

- Snap works with drag/keyboard

- Virtual desktops create

- Aero Peek functions

- Performance smooth

****Rollback:****

- Restore previous settings

- Document changes

- Note policy changes

****Escalation:****

- Graphics issues → Hardware team
- Policy questions → Security team
- Multiple users → Infrastructure

45. Windows File Association Issues

****Symptoms:****

- Wrong program opens files
- "How do you want to open this file?"
- Cannot change default programs
- File icons wrong

****Root Causes:****

- Program installation changed associations
- Registry corruption
- User profile issues
- Windows updates reset
- Third-party software conflicts

****Fix Steps:****

1. ****Right-click file**** > Open with > Choose default
2. ****Set defaults**** in Settings > Apps
3. ****Control Panel**** > Default Programs
4. ****Check file extension**** is correct
5. ****Reinstall affected program****
6. ****Reset all defaults**** if widespread

7. ****Check registry**** for file associations

8. ****Create new user profile**** to test

9. ****Run System File Checker**** (sfc /scannow)

10. ****Check for malware**** changing associations

****Verification:****

- Correct programs open files

- Icons show properly

- Settings persist

- All file types work

****Rollback:****

- Document previous associations

- Restore from backup if available

- Note program changes

****Escalation:****

- Multiple file types → Desktop team
- Registry corruption → Desktop team
- Malware suspected → Security team

Document Version Control

****Version:**** 2.0 - Enhanced with detailed fixes

****Last Updated:**** December 2025

****Next Review:**** March 2026

****Owner:**** IT Support Team

****Change Log:****

- **v2.0: Added comprehensive symptoms, root causes, detailed steps, verification, rollback, and escalation for all 45 runbooks**
- **v1.0: Initial generic template-based runbooks**

****Distribution:****

- **IT Service Desk (L1/L2)**
- **Desktop Support Team**
- **IT Operations Team**
- **Security Team (for security-related escalations)**